



Rock Field & Laboratory

Rock

Direct Shear

The Direct Shear System for Rocks contains closed-loop, servo-controlled, electro-hydraulic shear and normal load actuators which, in conjunction with the advanced software, allow for automatic performance of a Direct Shear Test. The software uses inputs from up to four normal deformation sensors and the normal load sensor to automatically calculate the average normal deformation of the specimen. This allows for the system to automatically perform either a constant normal stress or a constant normal stiffness test.



Portable Direct Shear For Rocks

This shear box has been designed to receive irregularly shaped samples with surfaces measuring up to 115 x 125 mm or cores with a maximum diameter of 102 mm. The lower half of the box is connected to two hydraulic rams which generate the shearing force (in both directions) and the upper half supports the ram that applies the load at a right angle to the failure surface. The horizontal and

vertical actuators are connected to two manually-operated hydraulic pumps. During the test the normal stress and the shearing stress are read on two independent dial gauges calibrated in engineering units; if the measurement of the shear deformation is required an appropriate dial gauge can be mounted on the support.



Triaxial Rock Testing Systems

The system is capable of performing static and dynamic closed-loop strain or stress controlled as well as post failure behavior tests. Included with this system is our true 32-bit Windows compatible software. It allows sending directly your test data to any computer connected to your local network.

Our new software includes a database system to keep track of all the data generated at your lab with any of our testing systems. Using a Windows network system in your lab facilitates the transferring of your test data directly into other Windows programs such as Word or Excel for report generation as well as to easily backup your important test results.

The system meets the specifications of the International Society for Rock Mechanics (ISRM) for triaxial testing of rock samples.



RAPID Triaxial Rock Testing System

In addition to the features of standard Triaxial Rock testing System offered by us, the Rapid Triaxial Rock Testing System is designed to perform rock triaxial tests simply and quickly.

An automatic hydraulic lift and sliding base for the triaxial cell simplifies specimen preparation and advanced software allows for a completely automated testing procedure.

It features a closed-loop, digitally servo controlled axial actuator for performing accurate triaxial and unconfined compression tests on rock specimens. Meets all ASTM and ISRM specifications.

In addition, the system is also capable of performing permeability, hydraulic fracturing, indirect tension, and many other advanced rock tests.



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Combined Direct Shear & Triaxial System

This system features electro-hydraulic closed-loop digital servo control of the shear and normal loads for test automation. The included software program accepts inputs from the normal load sensor and up to three normal deformation sensors (the software calculates automatically the average normal deformation). Loads or deformations for both the shear and normal actuators can be set to automatically perform advanced tests such as the Constant or Calculated Normal Stiffness Direct Shear Test.



Polyaxial/ True Triaxial Test

The Polyaxial Rock Testing system can be used to study the intermediate principal stress effect on rock engineering behaviour and describe adequately the strength of rock under a general system of Polyaxial compressive stresses. The system is designed to apply 3 independent stresses (σ_1 , σ_2 and σ_3) onto a cubical test specimen. The stresses are achieved through hydraulic intensifiers applying pressure to solid platens in contact with the specimen. The system can be upgraded with pore pressure and temperature control.

We also offer hydraulic fracturing fixtures to approximate the effects of anisotropic stress state on borehole stability. Ultrasonic sensors can also be provided to study the effects anisotropic stress conditions on compression and shear wave velocities. Velocities for all three axes can easily be measured during Polyaxial tests with this option.

Uniaxial Testing System

The Uniaxial Testing System is capable of performing static or dynamic uniaxial tension and compression tests as well as numerous other tests that can be used to determine the strength parameters of a rock specimen. The system features a closed-loop, electro-hydraulic servo, which, in conjunction with advanced software, can completely automate the testing procedure.



Hydraulic Fracture Test

The Hydraulic Fracturing Tester can be used to perform fracture tests, well bore stability tests, and permeability tests with or without a confining pressure. These tests can be performed at various temperatures, as an optional temperature control device can control the temperature up to 200°C. This makes the Hydraulic Fracturing Tester ideal for testing a wide variety of in-situ rock conditions.

The system includes a high-pressure triaxial cell and a hydraulic balance is used to apply either a hydrostatic stress state or apply a deviator stress in the vertical direction. It contains numerous feed-through lines for electronic measuring equipment and multiple fluid ports.

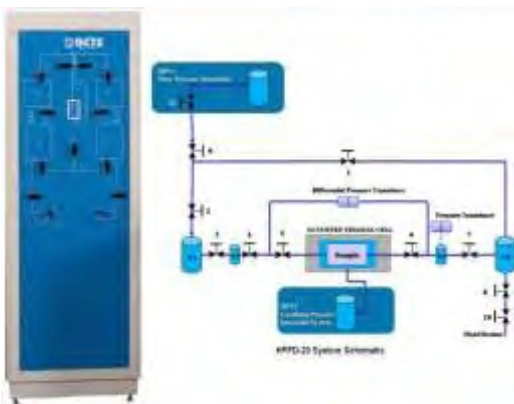


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Rock Creep Testing System

The Rock Creep Testing System is engineered to perform long term tests without the need of a continuous power input. This system is capable of operating without power for several days by using a compressor with a large tank with data acquisition being powered through the use of a battery. This design therefore eliminate the need for a power supply such as a power cord and socket that may limit the user. This system operates with only 800 kPa air pressure with minimum flow. The system includes load and deformation digital indicators along with an analog gauge which allows the user to easily read the applied load. Advanced software and USB based data acquisition are also included in order to simplify the gathering and recording of data and easily interface with the user's laptop/computer. This is therefore an ideal system for long-term testing and simple data gathering, thus enhancing the user's experience.

On-specimen deformation instrumentation can also be provided as well as triaxial cells and other testing components.



Fast Pulse-Decay Permeability Apparatus

The Fast Pulse-Decay Permeability Apparatus is designed for measurement of permeability in micro-porous material such as gas shale and other reservoir rocks, in order to determine the capacity and flow characteristics of the rock matrix. The system is capable of measuring rock permeability less than 1 μ D (microdarcy).

The system includes two stainless steel large reservoirs and two smaller stainless steel reservoirs. Also included are nine zero-volume-change ball valves and one needle valve for precise flow control.

The system components are secured in an insulated metal cabined with front panel mounted valves and controls. The system also features precise temperature control inside the cabinet for elimination of any errors from temperature induced volume expansion and contraction of the fluid and gas inside the reservoirs.

Includes all necessary valves and fluid lines to perform the Fast Pulse-Decay permeability test when used with any of our triaxial systems.

Ultrasonic Velocity Test System

The Ultrasonic Velocity Test System is a turnkey system and includes everything required to perform Ultrasonic Velocity measurements on laboratory specimens. The new hardware and software utilize the latest technology to allow for precise determination of wave velocities.

With the system, the user has the ability to control the pulse signal polarity and number of pulses to stack, this is important in helping to eliminate the influence of the Compressional wave signals on the relatively weaker Shear wave signals. The uses a fast-acting pulser that provides excitation to the ultrasonic sensor and an ultra high speed Analog-to-digital converter for storing the resulting waveforms signals. The sampling rate can be selected from 40 MHz to a sampling rate as low as 156 Hz allowing the user to capture a wide range of ultrasonic signals. Within the Ultrasonic software, the user is able to digitally control the receiver and pulser ensuring an easy setup and high degree of repeatability.

This system is provided with software that allows for automatic measurement while performing triaxial or unconfined shear loading. The ultrasonic data can be collected at any specified interval, such as time, stress, strain or any other test parameter.



Circumferential Velocity Anisotropy Apparatus

The Circumferential Velocity Anisotropy Apparatus is used to determine the compression P-wave velocities and shear S-wave velocities in rock core specimens in different orientations.

This apparatus, when used with our ultrasonic measurement device, can obtain the velocity versus angle and height data with relative ease and precision.





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Pore Fluid Transfer Cell

The Pore Fluid Transfer Cell when used with HPVC pressure intensifiers are ideal for the control of the pore pressure in triaxial tests, head pressure in permeability tests, or fluid pressure in hydro-fracture tests when corrosive fluids, such as brine, are used as the pore fluid. Transfer cell has a 400 cc volume capacity with absolute pressure of 140 MPa and differential pressure across the rolling diaphragm of up to 1,000 kPa. Larger Capacity cells are also available on request.



Hoeck Cells For Triaxial Tests

For cylindrical rock samples, cell length is twice the diameter. The cells are complete with caps, sleeve, high pressure joints and pistons. A kit of 5-6 sleeves and the couple of platens (load spreaders) are suggested



Automatic Pressurizer For Hoeck Cell

For use in combination with compression testing machine for performing tests with Hoeck cells (pressurizes automatically up to 700 bar). The console houses the power unit and digital control unit which acts as user interface. The system comprises Automatic, electro-hydraulic power unit characterized by very low noise and heat generation so it is ideal for continual usage. Monotronic digital display/readout unit with relevant strain transducer for measuring pressures up to 700 bar and Hydraulic hose with quick coupling.



Rock Deformation Device

The Rock Deformation Device measures axial and lateral strain directly on the specimen providing accurate results. The device consists of an upper and lower support ring machined with a low weight, high strength design.

This device can be used within our high-pressure triaxial cell and also used for elevated or cold temperature testing. Elastic constants such as Young's modulus and Poisson's ratio can be determined as well as inelastic strains that occur after sample failure (post failure analysis).

Direct servo control of axial or radial strains can be performed with this device.

Diametral Deformation Device

The Diametral Deformation Device is used to measure specimen diameter changes in order to calculate radial strains. This device mounts directly onto a specimen through the use of long screws attached to the arms of the device that press onto the specimen. By using these long screws, the device can be quickly and easily adjusted to fit a large range of specimen diameters.





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Differential Strain Curve Analysis Measurement Device

The Differential Strain Curve Analysis Measurement Device is designed for testing cubical rock specimens under hydrostatic conditions in order to determine the in-situ stress state. The results obtained using the device allow for characterization of the distribution of crack porosity with crack closure pressure as well as the crack orientation as a function of crack closure pressure amongst other parameters.



Indirect Tension/ Brazilian Test

The Indirect Tension Module is used to determine the indirect tensile strength by the Brazilian test. This device conforms to the test standard established by the International Society for Rock Mechanics (ISRM). The device has an upper and lower support ring created with a low weight, high strength design.



Rock Fracture Toughness fixture

The Rock Fracture Toughness fixture is designed to test cylindrical specimens under unconfined conditions to determine the fracture toughness of rock materials. The fracture toughness results allow for classification and characterization of intact rock with respect to its resistance to crack propagation. Fracture toughness data can be used in analysis of hydraulic fracturing (fracking) or as an index for rock fragmentation processes such as crushing or tunnel boring.



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Rock Abrasiveness Apparatus

The Rock Abrasiveness Apparatus is used for measurement of rock abrasiveness under the standards specified by the Cerchar test. The test consists of measuring the wear flat on the standard steel indenter with 200 kg/mm² hardness loaded with 70 N force, after it has scratched 10 mm distance on the rock surface. The indenter is examined under the microscope and the amount of wear is correlated to the Cerchar Abrasiveness Index (CAI).

The apparatus features precision slide for smooth movement of the rock specimen over the required scratch distance. It also comes with graduated knob for accurate scratch distance control with 0.01 mm precision. The slide movement is controlled with precision threaded acme rod with 1 mm pitch.



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Pressure Controlled Coring Machine

The Pressure Controlled Coring Machine features a pressure controlled feeding mechanism for even sample coring and it is able to achieve a large range of spindle speeds to provide optimum performance when preparing test specimens, regardless of the properties of the material being sampled.



Automatic Rock Lab Saw

Automatic rock core saw with 1 HP motor, 14" aluminum wheels regulate band-saw motion. Band-saw with industrial diamonds.



Specimen Grinder/ Computer Controlled Specimen Grinder

The Specimen Grinder provides the final step to preparing test specimens with parallel and flat ends according to ASTM and ISRM specifications. The grinder is driven by a heavy-duty 2.0 HP electric motor for durability. The grinder comes with a 177 mm diamond cup wheel, which can easily grind specimens from 25 mm to 150 mm diameter in a single pass.

The grinder can be supplied with the optional air cooling and dust systems. Also available is the optional automation upgrade for computer controlled grinding.

Slake Durability Apparatus

This apparatus assists in the evaluation of the resistance of rocks to disintegration when subjected to different drying and water-immersion cycles. The test consists in oven-drying a number of pieces of the material and then subjecting them to wear inside a drum rotating in water. This procedure is repeated two or more times; the disintegration index is determined by the percentage loss in weight that the sample undergoes.



Down-hole/ Seismic Refraction System

The Down-hole/ Seismic Refraction System is a PC based instrument that is very simple to operate. Some features include advanced Digital Signal Processing and signal enhancement including waveform stacking, filtering, and spectral analysis in real time.

The software package offers a convenient graphical user interface to measure P-and S-wave travel times and automatically calculate wave velocities. More advanced analyses are also possible with the use of sophisticated digital filters and spectral parameters that can easily be obtained with a click of the mouse. Special techniques are used in this program to correct or eliminate time shifting of the filtered waveforms. Acquired data may also be transferred to spreadsheet and database programs. 16-bit resolution coupled with state of the art software analysis tools provide for an ideal Down-hole or Seismic Refraction test system to measure in-situ compressional and shear wave velocities.

Designed for 6 in to 12 in diameter bore hole sizes, but optional accessories are available to accommodate smaller or larger boreholes.



Ground Penetrating Radars (GPR)

GPR

What is a GPR

Ground-penetrating radar (GPR) is a geophysical technique that uses radar pulses to image the subsurface.

How GPR Works

GPR works by sending a radio signal or series of signals into the ground. The returning signals provide information about changing ground characteristics with depth. Basic radar output presents this as patterns of signals at depth against the distance travelled along the ground by the radar. The radar measures depth in terms of the time it takes for a signal to return after emission. This makes GPR a very good detector of pipes & services (intrusive material), buried building materials (also intrusive) and changes in stratigraphy whether man-made (e.g. pavement profiling) or natural (e.g. peat basin detection). GPR can also be used to look inside structures either to check on the state of construction or to locate hidden objects (e.g. voids).

Why GroundVue ?

Groundvue range is designed by radar experts. They are the result of 30 years experience in advanced radar design – not only in Ground Penetrating Radar but also with avionics, military and SAR applications. Groundvue radars give you clear data quickly. They cut down on processing time: hardware design handles most of the clutter. Don't take our word for it – ask to see sample data.

Efficiency, Versatility and Toughness

These are the fastest GPR systems on the planet, transmitting simultaneously on all channels, recording positional data (GPS or Total Station) at the same time as the GPR data, allowing for fast efficient data collection. They are built to deal with a variety of environments.

Data Clarity

Clarity really matters. It's what allows you to take decisions on-site. It can make the difference between detection and non-detection, particularly in difficult environments. It can also save you time in post-processing.



User Friendliness

A simple user interface which is easy to operate. Suited to operation by individuals whose expertise and technical abilities are drawn from their own disciplines - not necessarily GPR.

Data is always recorded unprocessed. Real time viewing of the data can be in raw or processed format, but data recorded in raw format minimises the risk of inappropriately recorded data (or, worse, the need to redo a survey).

Regulation Compliance

Groundvue radars are designed to meet the ETSI specifications, EN-302 489 32. Can also meet FCC regulations if required.

Award Winning Technology

Last, but not the least, as well as the Groundvue GPRs, Utsi Electronics Ltd carry out a number of specialist designs for clients. Most of the Groundvue range of GPRs originated in this sort of development work.



Ground Penetrating Radars (GPR)

The GroundVue Range

Utsi Electronics offers, what can be considered as one of the widest range of GPRs, from extremely shallow depth scanning of few centimeters to hundreds of meters.

Table:

FREQUENCY	DEPTH	CHANNEL	CONDITIONS	RADAR
6GHz	0 to 400mm	Single	Dry	Groundvue 5C
4GHz	0 to 0.5m	Single	Dry	Groundvue 5
1.5GHz	0 to 1m	Up to 4	Dry	Groundvue 3
1.5GHz	0 to 1m	Single	Inside ducting	Borevue
1GHz	0 to 1.5m	Single,up to 4	Dry	Groundvue 3
200 to 600MHz	0 to 5m	Single,up to 4	Dry	Groundvue 3
200 to 600MHz	0 to 3m	Single,up to 4	Wetland	Groundvue 3
125 to 500MHz	0 to 10m	Single,up to 4	Dry	Groundvue 3
125 to 500MHz	0 to 5m	Single,up to 4	Wetland	Groundvue 3
30 to 100MHz	0.5 to 20m	Single	Dry	Groundvue 2
30 to 100MHz	0.5 to 11m	Single	Wetland	Groundvue 2
30 to 100MHz	0.5 to 11m	Single	Freshwater	Watervue
10 to 30MHz	0 to 200m	Single	Any	Groundvue 6
40MHz	0 to 100m	Single	Any	Groundvue 7

Groundvue 2

Groundvue 2 is a user friendly and uses easy to operate dedicated software. Operating depth (measured in travel time) and the distance between samples must be defined accurately. All other parameters can be reset after survey.

Typical Applications

For environmental monitoring including wetland & peat, and archaeological detection in wetlands.

Depth Range

At 640ns:

0.5m to 30m in dry conditions

0.5m to 11m in waterlogging

At 1280ns:

0.5m to 60m in dry conditions

0.5m to 20m in waterlogging

Resolution : 200mm

Special Features

Arrayed antennas, fully screened, suitable for towing behind a vehicle during survey. Optional GPS/Total Station data input. Data storage direct to solid disk



Ground Penetrating Radars (GPR)

GroundVue 3_8 Channel

This is the latest high speed flexible Groundvue multi-channel. It can be used in any configuration, from a trolley, from a vehicle, hand towed. Antennas can be attached, detached for another application, changed around at will. And the viewing parameters can be changed during operation. This is the radar used to detect fibre optic cables (video coming soon) and also to accurately track the direction of utilities at one sweep.

Typical Applications

The wide range of antenna frequency makes this an extremely versatile equipment. The applications range from scanning for structural integrity of RCC structures, rebar mapping, fire damage, road pavement inspection, utility mapping, archeological surveys etc. to name a few.

Antenna Frequencies

250MHz, 400MHz, 1GHz, 1.5GHz, 4GHz, 6GHz. 100MHz coming soon.

High Speed Multi-Channel

- Up to 1,600 scans per second, data collection speed consistent with traffic speeds
- Simultaneous triggering without cross channel interference
- Up to 8 channels: select any number from 1 to 8



Wi-Fi

- Wi-Fi operation, Ethernet cable for frequencies > 1GHz where Wi-Fi cannot be used

Crack Depth Measurement

Any GPR can detect areas of generalised cracking but the special crack detection head is used for:

- Accurate detection of joints in concrete
- Depth measurement of surface (top down) cracking
- Detection of subsurface (bottom up) cracking on roads, airports and in structures such as concrete loading bays and flat roofs

Automatic Velocity Calibration

- Groundvue antennas are simultaneously triggered, not sequentially, and without cross channel interference. Using multiple receivers connected to a single transmitter, continuous comparative data is created. Layer tracing generates the information needed to calculate transmission velocity for the full length of the data collected. In the process, any anomalous areas show clearly.



Groundvue 3_1

This is the single channel version of the wireless Groundvue 3_8 and can be used with any of the Groundvue 3 antennas (one at a time).

Ground Penetrating Radars (GPR)

Groundvue 5

Groundvue 5 is user friendly and uses easy to operate dedicated software. Frequency ranges are 2GHz to 6GHz .

Special Features:

- Hand held (dimensions excluding connectors and handles: 240x16x90mm)
- Lightweight
- Horn antennas
- Fully screened
- Automatic Mapping Programme available (at no extra cost)

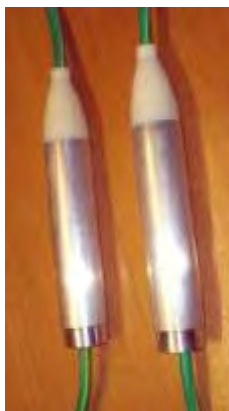
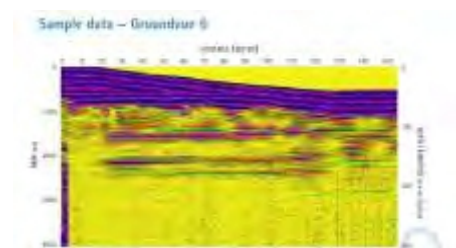


Groundvue 5C

Frequency Range is 6 GHz central (1GHz to 8GHz) and Depth Range Up to 400mm.

Special Features:

- Highest available GPR resolution capability
- Handheld
- Lightweight
- Dimensions excluding connectors and handles: 240x16x90mm
- Horn antennas.



Groundvue 6

Groundvue 6 is a thin inline spread spectrum GPR pulled behind the operator for harsh environments. Frequency Range is 7MHz to 30MHz & Depth Range is Up to 180m.

Special Features:

- Hermetically sealed
- Internal batteries
- Wireless synchronization and data transfer
- Ultrasensitive correlation receiver
- Internal data stacking (to improve signal to noise ratio)
- GPS interface

Groundvue 6C

Frequency Range 15MHz to 60MHz and Depth Range Up to 90m.

Special Features:

- A thin inline spread spectrum GPR pulled behind the operator for harsh environments
- Hermetically sealed
- Internal batteries
- Wireless synchronization and data transfer
- Ultrasensitive correlation receiver
- Internal data stacking (to improve signal to noise ratio)
- GPS interface



Groundvue 7

This low frequency (30MHz) GPR with snake antennas was formerly subject to an exclusivity agreement. As this agreement has now been cancelled, this radar is now available for hire or purchase.

